

(12)

## SIMATS ENGINEERING ASSESSMENT TOOL 3

**COURSE NAME: COMPUTER NETWORKS**  
**COURSE CODE: CSA07**

### COURSE OUTCOME COVERED

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

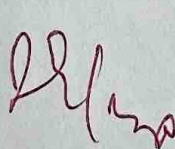
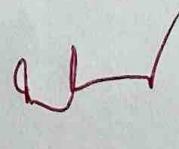
I M. Farah Binti / 192511434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Farah Binti

Signature of the student:

| Criteria               | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks | Q5 10marks |
|------------------------|-----------|------------|------------|------------|------------|------------|
| Problem Identification | 20%       | 1.4        | 1          | 1          | 1          | 1          |
| Debugging              | 25%       | 1.6        | 1          | 1          | 1          | 1.6        |
| Optimization           | 20%       | 1          | 1.4        | 1.4        | 1.6        | 1          |
| Analysis               | 20%       | 1          | 1          | 1          | 1          | 0.4        |
| Code Quality           | 15%       | 1          | 1.6        | 1.6        | 1.4        | 1          |

CO-2

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## Assessment Tool-3

### Debugging and Optimisation Task

1) Identify the subnet of the Configured IP Address

| Subnet            | Address Range                   |
|-------------------|---------------------------------|
| 192.168.10.0/26   | 192.168.10.0 - 192.168.10.63    |
| 192.168.10.64/26  | 192.168.10.64 - 192.168.10.127  |
| 192.168.10.128/26 | 192.168.10.128 - 192.168.10.191 |
| 192.168.10.192/26 | 192.168.10.192 - 192.168.10.255 |

The workstation IP: 192.168.10.65 belongs to subnet:

192.168.10.64/26

(i) Network Address Broadcast Address, and Host Range

192.168.10.64/26:

| Parameter         | Address        |
|-------------------|----------------|
| Network Address   | 192.168.10.64  |
| First valid host  | 192.168.10.65  |
| Last valid host   | 192.168.10.126 |
| Broadcast Address | 192.168.10.127 |

### 3. Verification

#### IP address Verification

192.168.10.65  $\rightarrow$  belong to 192.168.10.64/26

$\downarrow$  valid

IP address is valid

#### Gate way Verification

Configured gateway: 192.168.10.1  $\xrightarrow{\text{belongs to}}$  192.168.10.0/26

$\downarrow$  Host range

Gateway is not valid  $\leftarrow$  192.168.10.1 - 192.168.10.62

### iv) Configuration errors

Error 1: wrong subnet

$\rightarrow$  IP address 192.168.10.65 belongs to 192.168.10.64/26

$\rightarrow$  Other systems are 192.168.10.0/26

Error 2: wrong gateway

$\rightarrow$  Gateway 192.168.10.1 belongs to 192.168.10.0/26

$\rightarrow$  The IP and gateway are not in the same subnet

### v) Correct configuration

| setting         | value                 |
|-----------------|-----------------------|
| IP address      | 192.168.10.2          |
| Subnet mask     | 255.255.255.192 (/26) |
| Default Gateway | 192.168.10.1          |

## Usable host calculation

$$\text{Total address} = 2^{32-26} = 64$$

$$\text{Reserved Address} = \text{Network} + \text{Broadcast} = 2$$

$$\begin{aligned}\text{Usable Hosts} &= 64 - 2 = 62 \\ &= 62\end{aligned}$$

② Given: i) Expanded IPv6 Addresses

Device A : 2001:0db8:0000:0000:0000:8800:042:329 /64

Device B : 2001:0db8:0000:0000:0001:0000:0000:0001 /64

ii) Network Prefix

For a /64 Prefix, the first four hexets represent the network

Device A Prefix:

2001:0db8:0000:0000:: /64

Device B Prefix:

2001:0db8:0000:0000:: /64

### iii) Verify same Subnet

Both devices have the same network prefix.

2001:db8::0:0::/64

Both belongs to the same subnet

### iv) Address / Prefix mismatch

\* There is no prefix mismatch

\* Device A and B are correctly configured in the same /64 subnet

\* Communication failure occur due to:

↳ Incorrect IPv6 configuration

↳ Disabled IPv6 interface

↳ Firewall settings

↳ Route

### iv) Address / Prefix mismatch

| Device   | IPv6 Address        |
|----------|---------------------|
| Device A | 2001:db8:0:0::10/64 |
| Device B | 2001:db8:0:0::20/64 |

Both Belongs to

Network Prefix: 2001:db8:0:0::/64

## usable Host Addresses

$$\rightarrow \text{Host bits} = 128 - 64 = 64 \text{ bits}$$

$$\rightarrow \text{Total host addresses} = 2^{64}$$

$$2^{64} = 18,446,744,073,709,551,616$$

③

### Subnet details:

#### Subnet 1: /26

\* Network Address : 192.168.1.0

\* Broadcast Address : 192.168.1.63

\* Host Range : 192.168.1.1 - 192.168.1.62

\* Usable Host : 62

#### Subnet 2: /27

\* Network Address : 192.168.1.64

\* Broadcast Address : ~~192.168.1.95~~

\* Host Range : 192.168.1.65 - 192.168.1.94

\* Usable Hosts : 30

#### Subnet 3: /28

\* Network Address : 192.168.1.96

\* Broadcast Address : 192.168.1.111

\* Host Range : 192.168.1.97 - 192.168.1.110

\* Usable host : 14

## ii) Usable and Unused IP Addresses

| Subnet | Total IPs | Usable hosts | Unused hosts                       |
|--------|-----------|--------------|------------------------------------|
| 126    | 64        | 62           | Depends on department requirements |
| 127    | 32        | 30           | Depends on department requirements |
| 128    | 16        | 14           | Depends on department requirements |

## iii) Inefficient Allocation:

- 126 subnet provides 62 hosts. If the department has fewer devices, many addresses remain unused
- 128 subnet provides only 14 hosts, causing shortage if more devices are required
- Fixed subnetting wastes addresses because all departments do not require equal sizes

## iv) Addressing Plan errors

- \* Large departments may have been assigned smaller subnets (128) causing insufficient IPs.
- \* Small departments may have been assigned larger subnets (126), causing inefficient IP wastage
- \* Subnet sizes are not matched with actual device requirements

## Addressing / Prefix Mismatch

There is no prefix mismatch. Both devices are in the same /64 subnet. Communication failure may due to other issues (eg firewall, interface or routing)

### VI) optimized IPv6 configuration

| Device   | IPv6 Address        |
|----------|---------------------|
| Device A | 2001:db8:0:0::10/64 |
| Device B | 2001:db8:0:0::20/64 |

Both devices remain in the subnet 2001:db8:0:0::/64

### VII) Available host addresses

→ Host bits = 64

→ Total available host addresses =  $2^{64}$   
= 18,446,744,073,709,551,616

### 5) 1) Matching routing entry

The destination IP 192.168.2.100 belongs to:

192.168.2.0/24

Matching route: 192.168.2.0/24

## V) Optimized VLSM scheme

| Department requirement | Recommended subnet | Hosts    |
|------------------------|--------------------|----------|
| Large department       | /26                | 62 hosts |
| Medium department      | /27                | 30 hosts |
| Small department       | /28                | 14 hosts |

## 4) i) Expand the IPv6 Addresses

Device A:

2001:0db8:0000:0000:0000:0000:0042:8329 /64

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001 /64

## ii) Network Prefix

For a /64 Prefix, the network prefix is the first 64 bits

Device A: 2001:0db8:0000:0000:: /64

Device B: 2001:0db8:0000:0000:: /64

## iii) Do both the devices belong to the same subnet?

Yes both the devices have the same network prefix!

2001:db8:0:: /64

## next-hop router

The packet will be forwarded to:

Next-hop: 10.0.0.3

### iii) Verify Destination Subnet

→ Destination IP: 192.168.2.100

→ Network: 192.168.2.0/24

### iv) Routing Configuration

\* The routing table entry is correct

\* If packets are not reaching the destination, issues are:

↳ Incorrect Next-hop configuration (10.0.0.3)

↳ Interface/link failure

↳ Destination network is unreachable

### v) Optimized Routing table

| Destination network       | Next hop |
|---------------------------|----------|
| 192.168.1.0/24            |          |
| 192.168.2.0/24            |          |
| Default Route (0.0.0.0/0) |          |

vi) Route used if no match exists

Default route → 0.0.0.0/0 via 0.0.0.1